

SMPC PLANT INTERLOCK (PTA UNIT)

PTA No.1

WIS-2101 HIGH WEIGHT (SD-2101)

When the CTA feed is stopped, PTK-101 becomes full unless receiving of CTA is stopped. Therefore, in the case of PTK-101 excess weight, the equipment on the upstream side is stopped.

CTA FEED system STOP (SD-2101)

If the equipment on the downstream side is stopped in the CTA feed mechanism, CTA deposits and clogging occurs, so the equipment on the upstream side is stopped.

PP-102A,B,C STOP (SD-2102)

If all the three units of PP-102A,B and C are stopped, the PP-102 discharge shut-off valve is closed to prevent back flow from PD-201. further, TV-2101 for the hot oil which is a heating source of PE-101E and F is closed, the PF-601 main burners (A and C) are extinguished, and the flow rate of the PE-601 hot oil is raised to the setting to cover the heat load

PP-102A,B,C Starting Conditions (SD-2103)

PP-102A, B and C cannot be started (PS-2108) unless the pressure of the lubricating oil is 1.2 kg/cm²G or more

REACTOR HIGH PRESS (SD-2201)

If clogging or a blow-down failure occur in the reactor, the internal pressure of the reactor rises. In order to prevent complete blocking, if REACTOR HIGH PRESS occurs, the recycle solvent operation is automatically started to prevent feed or reactor clogging. If the slurry in the reactor needs to be displayed immediately due to trouble in a different system, such a SW is installed as is automatically placed in RS operation whenever necessary.

AGITATOR SEAL WATER LOW PRESS (SD-2301)

DIW pressurized by PP-504 is introduced to the mechanical seal section of each agitator at the specified flow rate. Each seal pressure is controlled by PICS to the drum pressure – 5kg/cm²G.

If PICS falls below the pressure setting due to a certain trouble, SDV is closed and the accumulator is activated to keep the pressure of the seal section. When the pressure is recovered, SDV is opened.

PM-401A,B SD (SD-2401A/B)

If the seal coolant pressure, the spindle pressure or the gear box pressure falls below the setting and the torque limiter is activated, the centrifuge is stopped and the feed LICV is closed to prevent damage to the centrifuge.

If the oil pump stops, the oil return SDV is closed and the oil pressure is kept by the accumulator.

PM-401D SD (SD-2401D)

When pressure or Temp feed line of PM-401D high more than setting feed valve will close prevent damage of PM-401D

When pressure seal C.H , Seal drum diff less than 0.1K/G or torque high than setting PM-401D will stop and valve feed line will close prevent leak pass zone

When re slurry water low flow <10 T/H feed valve will close prevent bottom line plug

PP-502A/B STOP (SD-2502)

When PP-502A and B stop slurry feed to PM-401A/B will close prevent bottom of PM-401A/B plug

PM-402A/B HIGH LEVEL (SD-2402A/B)

If the overflow line from PM-402 to PD-402 is clogged and the level in the pan rises, there is a possibility of inflow of the slurry to the dryer side. To prevent this, a level gauge (LIS) is provided and the feed to PM-402 is stopped in response to HIGH LEVEL.

In addition, the sequence for the filter fabric washing LPW in operation is stopped to prevent inflow.

PD-402 HIGH PRESS, HIGH LEVEL (SD-2410)

If the pressure of the PD-402 vent system rises, the PD-402 slurry flows backward in the PM-402 overflow in line and overflow to the dry side. If the PD-402 level rises, the PD-402 vent line is clogged up due to entrainment of TA. To prevent these accidents, LV-2401 is closed if the pressure of P-2406 becomes high ($0.25 \text{ kg/cm}^2\text{G}$) or the level of PD-402 becomes high (45%)

PM-402 HIGH & LOW PRESS (SD-2420A/B)

When PM-402 is maintained or repaired, it is separated from the dryer system. If PM-402 HIGH or LOW PRESS occurs, SDV is opened to prevent damage to the casing caused by excess pressure due to gland N_2 or a negative pressure due to leak from the filtrate KCV.

PD-405 HIGH LEVEL (SD-2403)

If the level of PD-405 become high due to trouble in PP-405 or clogging of the SUC&DEL line, the PM-402 filtrate suction line is clogged up with the filtrate, resulting in the lowered flow rate of the blow back gas or the clogged tube with PE-405 TA.

Therefore, if the PD-405 level exceeds the setting, the PM-402 vacuum is cut and the RVF treatment is stopped.

DRYER SD (SD-2404)

In the case of trouble in the related equipment of the dryer system, if any equipment is stopped due to a certain cause, all the equipment on the upstream side of the relevant equipment is stopped to prevent deterioration of the product quality due to damage to the equipment, clogging or a drying failure.

If the flow rate of the pneumatic N_2 or the drying N_2 lowers or if the level of PTK-800A, B or C becomes very high, the dryer is shut down and the pneumatic transfer is stopped.

For RVF, the RVF ESD SW is set to ON and the treatment is stopped.

PT-401 HIGH LEVL (SD-2405)

If the PT-401 level rises due to clogging of the PP-410 SUC&DEL line or a pump breakdown, the water may flow to the dryer side. To prevent this, the PT-401 top spray water is stopped if the level of PT-401 becomes high.

When PT-401 returns to the normal level (50%), the interlock is automatically released and the top spray water is placed in auto control.

PC-402 STOP (SD-2406)

When PC-402 is run and the dryer outlet TA is pneumatically transferred to the product silo (PTK-800), there is a possibility that the product silo will be placed under a negative pressure. The negative pressure in the silo causes damage to the silo and excess pressure of the PC-402 suction cause damage to PC-402.

Therefore, if the pressure of the PC-402 suction become high or low, PC-402 is stopped.

PP-450 high flow (SD-2450)

When FI-2450 high flow or recycle slurry or recycle solvent PP-450 will stop

PTA No.3

PC-402 STARTING CONDITION (SD-2406)

PC-402 can be started only when KCV-2804A (to atmosphere) at the PM-803 outlet is closed and KCV-2804B at the PC-402 suction is opened, or when KCV-2801 is set to the PTK-800 side.

PD-501 HIGH LEVEL (SD-2501)

If the level of PD-501 becomes high, receiving of LPW to PD-501 is stopped (LIC-2501B closed) and the recovery of RS to PT-501 is stopped (LICV-2501A closed).

PT-501 SD (SD-2510)

PT-501 is a vent scrubber of the crystallizer system. In order to prevent damage to the equipment, if the pressure rises to the setting (1,000 mmH₂O), the PE-501 bypass valve is opened to prevent excess pressure of PT-501. when the pressure is recovered (500 mmH₂O), the PE-501 bypass valve is closed.

If the internal pressure of PT-501 drops, there is a possibility that the air is sucked from the PZ-501 vent. As H₂ gas exists in this line, the 3S valve is opened if the internal pressure lowers (pressure setting : 100 mmH₂O), in order to prevent dilution of the H₂ gas or suction of the air. When the internal pressure is recovered (pressure setting : 150 mmH₂O), the 3S valve is closed.

PE-601 SD (SD-2601)

If the flow rate of the main burner HO lowers (FICS-2601 low flow) due to trouble in the hot oil pump or the clogged SUC strainer, the equipment may be damage due to abnormal heating of the F-601 heating tube.

In addition, abnormal heating of the F-601 heating tube (TICS-2601 HIGH TEMP) and a drop in the PF-601 burner pressure (PIS-2611 LOW PRESS) will cause an accidental fire or explosion in the furnace. Therefore, if the above elements are activated, all the main and pilot burner are extinguished.

If PC-601 or PC-602 is stopped, the draft of PF-601 becomes difficult, so all the burners are extinguished. Note however that if both PC-601 and PC-602 are stopped, the snuffing steam is blown to PF-601 and purge is performed for 15 min.

PROTECT MISOPERATING FOR PTA SILO (SD-2801)

(1) KCV-2801, KCV-2802 and KC-2803 cannot be opened/closed by hand unless pressure equalization valves KCV-2808 and KCV-2809 are opened. KCV-2801 and KCV-2883 can be opened/closed if KCV-1402A is opened and KCV-1402B is closed (indicating that TC-302 is stopped).

(2) KCV-28014A/B can be opened/closed by hand if PC-402 is stopped. (These valves cannot be opened/closed if PC is in operation)

PTA SILO SD (SD-2802)

The operation prevention device is installed to prevent a rise in the internal pressure of the silo and to prevent suction of the air to the silo.

If the flow rate of the pneumatic conveying air becomes low, the pneumatic line will be clogged up.

Therefore, if the flow rate of FICS-2800 falls to the setting, KCV-2801A/B/C is closed and PZ-800A/B/C is stopped.

PP-601A,B,C high temp

When temp of motor PP-601A,B,C high than 180°C PP-601A,B,C will stop prevent motor coil damage

PP-601A,B,C Auto start

When PP-601 one unit stop ,Stand by unit will automatic start prevent main process PE-101F temp low

PTA No.4

PROTECT MISOPERATING FOR PTA PNEUMATIC LINE (SD-2811)

The preventive device against operation errors is installed to prevent the operation of KCV in error while the product PTA is being pneumatically transferred to the packing silo.

KCV-2810, KCV-2820, KCV-2833 and KCV-2885 can be operated and the direction can be switched if PZ-800A,B and C are stopped (pneumatic transfer being stopped). However, if any of PZ-800 is in operation, these valves cannot be operated.

PD-901 NaOH INFLUX PROTECTION (SD-2901)

If KCV-2910 is opened in error during PM-901 NaOH washing, NaOH enters PD-901. therefore, opening of KCV-2910 is disabled while the PM-901 NaOH washing sequence is in operation.

PZ-411 PRESSURIZING PROTECTION (SD-2411)

During PD-404 NaOH washing, blow is done from the PD-403 bottom to PZ-411. at this time, if KCV-2456 of the PZ-411 vent line is closed, PZ-411 may be damaged. Therefore, the preventive device against operation errors is installed to disable the operation of KCV-2456 when KCV-2453 used to blow from the PD-403 bottom to PZ-411 is opened.

KCV-2456 is opened according to the PD-404 NaOH washing sequence.